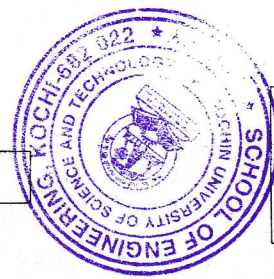


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**B.Tech. Degree III Semester Examination November 2018****CS/IT 15-1304 OBJECT ORIENTED PROGRAMMING**
(2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) What is the I/O function available for reading an entire line of text from console? Write the syntax.
- (b) What are reference variables? What is the situation where pointer variables are preferred over reference variables?
- (c) Explain the following OOPS features using real life examples.
- (i) Data Abstraction.
 - (ii) Polymorphism.
- (d) What is the memory layout structure for classes and objects?
- (e) Write the syntax of copy constructor. Mention two situation in which it is implicitly called.
- (f) What is the use of pure virtual functions? What is the effect it gives to the residing class?
- (g) What is 'this' pointer? Illustrate its use in an operator function used for operator overloading.
- (h) Give the general syntax of file open () command. What is the difference between ios :: app and ios :: ate mode?
- (i) Give any two functions available for writing blocks of characters to file.
- (j) What are the restrictions for writing destructor functions? When are destructor functions invoked implicitly?

PART B

(4 × 10 = 40)

- II. (a) What are inline functions? How its execution pattern is different from ordinary functions? (5)
- (b) The current passing through a circuit is given by $I = V/R$, where V is the applied voltage and R is the equivalent resistance. Write a function to evaluate the current when two resistances R1 and R2 are connected, either in series or parallel. If not specified, take the connection to be series. (5)

OR**(P.T.O.)**

- III. (a) How do a function call gets matched to a set of overloaded functions. (4)
 (b) Illustrate the following parameter passing techniques with suitable examples. (6)
 (i) Call by value.
 (ii) Call by address.
 (iii) Call by reference.

- IV. (a) List any four features of constructor functions. (4)
 (b) Define a class to store a rational number p/q. Overload the '+' operator so that we can add (6)
 $fr1 = fr1 + fr2$
 $fr1 = C + fr2$
 $fr1 = fr2 + C$ where fr1, fr2 are objects of type rational and C is a constant.

OR

- V. (a) What are static members of a class? Show the declaration and the usage of them. (4)
 (b) Define a class C string to store a character string and to provide the following operator functions. (6)
 == for equality
 + for concatenation and
 > for greater than.

- VI. (a) Differentiate between multilevel and multiple inheritance. How are constructors invoked in each case? (3)
 (b) Define a class to store the following information about an element. (7)
 (i) Symbol (ii) Atomic no. (iii) Atomic wt.
 Create a derived class to store the following additional information.
 (i) Number of protons (ii) Number of neutrons.

Write member functions to read and display the entire information about an element and an additional capability to compare the atomic number of two elements and return the symbol of the one for which atomic number is more.

OR

- VII. (a) How is run time polymorphism achieved with virtual functions? Explain the mechanism. (3)
 (b) Classes Time and Date are designed with its own member functions to initialize its own data member. Derive a new class 'clock' from both these classes which include a member function that will increment time and date. (7)

- VIII. (a) What are file pointers and the manipulating functions of corresponding file pointers? (4)
 (b) Write a program to create a text file and to later read it and count the number of vowels in it. (6)

OR

- IX. (a) Using an example show how exception handling statements can prevent 'divide by zero' exception condition. (4)
 (b) Write a program to create a binary file which contains objects of laptop with details such as model number, memory capacity, HDD size and price. The program searches the file with model number and display the details of laptop. (6)

